Public vessel-tracking data does not predict the Henry Hub–TTF spread: a pre-registered, power-bounded null

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Abstract

Positions of liquefied-natural-gas carriers are widely believed to carry information about the price gap between North American and European gas: a cargo that leaves the US Gulf today lands in Europe in two to three weeks, and that supply is not yet in any published statistic. We test the belief directly. From 54,494,686 AIS position fixes (2016-01-01 to 2026-08-10), reconstructed into 87,874 port events and a 34-signal daily panel built twice — once with hindsight for validation and once strictly point-in-time for modelling — we ask whether tanker-derived signals predict the one- and four-week change in the Henry Hub–TTF spread net of its own persistence, weather, storage and oil. Every hypothesis, sign and acceptance bar was committed to a dated append-only log before the corresponding fit. The answer is no. A controls-only model is worse than assuming no change (-9.3% at one week, -18.2% at four); no signal reaches a Newey–West $|t|$ of 1.96 (maximum 1.50); and three pre-registered mechanisms that could hide an effect from a linear scan — the Europe-bound *share* of cargo, an interaction with EU storage, and a tightness-regime split — fail under a Bonferroni bar with an untouched holdout. The design detects effects of at least 1.72% of residual variance and observed at most 0.009, so the result rules out tradeable effects and is silent on sub-tradeable ones. Alongside the null, we document two construction hazards specific to voyage-derived features: the unit of observation is defined retrospectively, and feature *availability* changes across data vintages. A naive loader leaked 65% of the sample.

1 Introduction

The Henry Hub–TTF spread is the price of moving a molecule of gas from the United States to Europe, and liquefied natural gas (LNG) carriers are how the molecule moves. A laden carrier leaving Sabine Pass arrives at Rotterdam roughly fourteen to eighteen days later. Between departure and arrival its cargo is committed, physically visible to anyone with a receiver, and absent from every published supply statistic. The inference that vessel positions must therefore lead the spread is intuitive, widely held, and commercially sold.

This paper tests it, and finds no support. The contribution is not the null itself — an efficient-markets reading predicts it — but the form in which the null is delivered. Three things make it a result rather than an absence:

1. **Pre-registration.** Every target, control set, lag, sign and acceptance bar was written into a dated, append-only decision log before the corresponding fit, so that no choice below was made after seeing what it produced. Where a pre-registered sign was falsified, it is reported as falsified.

2. **Inference that respects the data.** Weekly changes over a four-week horizon overlap; standard errors are Newey–West throughout, with the lag fixed in advance at $h+1$ weeks. Training rows whose target reached into the test week are purged. The three follow-up mechanisms are held to a Bonferroni bar and must replicate their sign on a holdout that was not examined until they had passed on discovery.

3. **A bound on what could have been found.** With 327 weekly observations the design detects effects of at least 1.72% of residual variance. The largest observed anywhere is 0.009. The result therefore rules out effects large enough to trade and says nothing about smaller ones.

Two further findings are methodological and, we think, of independent use to anyone building features from vessel-tracking data. First, the natural unit of observation — a *voyage* — does not exist until both its endpoints have been observed, so any statistic computed over completed voyages embeds the future: at a 2020 as-of date a straightforward loader returned 20,351 legs of which 13,130 (65%) had been closed by arrivals that had not yet happened. Second, feature *availability* changes across data vintages: the same loader attached 2,101 destination declarations to 2020 voyages from a data feed that did not exist until 2026, which no date filter on observations can catch.

The rest of the paper is organised as follows. Section 2 describes the pipeline and the decade-scale reconstruction. Section 3 sets out how the signal panel is built point-in-time. Section 4 describes the pre-registration protocol and its limits. Section 5 reports the physical nowcasts, which converge on the naive mean. Section 6 reports the spread tests. Section 7 bounds what they could have found. Sections 8 and 9 interpret and qualify.

2 Data and pipeline

2.1 Sources

Vessel positions come from three feeds with different fidelity, and the paper is careful never to aggregate across their seams. For the United States we use the NOAA Marine Cadastre archive of terrestrial Class-A AIS [10], which is exhaustive along the US coast from 2016 and ends on 2025-12-31. For Europe and the open ocean we use Global Fishing Watch port-visit and voyage events [5], which supply arrivals and departures but carry no coordinates and no anchorage events; that feed ends on 2026-02-21. A live subscription to an AIS aggregator covered a tier-ranked fleet from April 2026 and is used only for the vintage self-validation described in Section 3.

Prices and controls are all public: Henry Hub spot and Lower-48 storage from the EIA API [12]; the TTF front-month contract from a keyless quote feed, cross-checked against the World Bank's monthly European gas series [13] (slope 0.999, $R^2 = 0.9999$ over 106 months, so the continuous-roll distortion is negligible); EUR/USD and Brent from FRED [2]; EU storage from GIE AGSI+ [4]; and heating/cooling degree days computed from ERA5 reanalysis temperatures at weighted demand centres [11].

2.2 Reconstruction

Table 1 gives the scale of the reconstruction. Fixes are resolved against hand-drawn berth, anchorage and approach polygons at every LNG terminal in seven zones (US Gulf, US Atlantic, north-west Europe, Baltic, Iberia, western and eastern Mediterranean) by a per-vessel state machine that emits zone entry, anchorage, mooring and departure events. Three pairings of

Table 1: Reconstruction inventory. Port events are labelled by the feed that produced them; the two backfill feeds end months before the live feed, which matters for the sample choice in Section 6.

layer	rows	span
AIS position fixes	54,494,686	2016-01-01 to 2026-08-10
port events (state machine)	87,874	
noaa	44,363	to 2025-12-31
gfw	40,273	to 2026-02-21
mmsi_filter	1,974	to 2026-08-10
bbox	1,264	to 2026-05-30
in-scope fleet (LNG carriers + FSRUs)	828	
laden US export departures	9,855	
signal panel rows (`signal_daily`)	1,032,982	34 keys, two bases
model panel rows (`model_panel`)	77,500	20 features

Table 2: NOAA-only capture of US LNG export cargoes against the EIA-implied count. The shipped diagnostic tool counts every feed and so double-counts departures seen by both NOAA and GFW; the series here is the single-feed one.

year	captured	EIA-implied	capture
2016	48	51	95%
2017	159	192	83%
2018	216	294	74%
2019	290	494	59%
2020	300	648	46%
2021	715	966	74%
2022	918	1048	88%
2023	1,099	1178	93%
2024	1,216	1185	103%
2025	1,570	1494	105%

those events are the substrate for everything that follows: a *leg* pairs a departure with the vessel's next zone entry (a voyage), a *visit* pairs a mooring with the next departure (berth occupancy), and a *queue* pairs the first anchorage entry with the next mooring at the same terminal (the wait to berth). Laden state is inferred from reported draught against design draught.

2.3 Ground truth

The US leg of the reconstruction can be checked against a published total. The EIA reports monthly US LNG exports; dividing by a nominal 174,000 m^3 cargo gives an implied cargo count, and the ratio of laden departures we observe to that count is a capture rate (Table 2). Restricted to the NOAA feed alone it is 46% in 2020, rising to 93%, 103% and 105% in 2023–25. The gradient is a coverage artefact of the archive's early years, and it means early-year *levels* of any US signal are biased low by up to half; models below either work on within-window changes or standardise, and none reads an early-year level as a physical quantity.

Figure 1 shows the target and the headline signal on one timeline. The eye finds the two structural events every model must survive — the 2020–21 inversions in which Henry Hub briefly exceeded TTF, and the 2022 European crisis — and little else.

Figure 1: The Henry Hub–TTF spread (top) and the EU-bound gas at sea (bottom), daily, on the coverage-consistent sample used for modelling. The extremes land on Winter Storm Uri (February 2021) and the August 2022 TTF peak.

3 Building the panel point-in-time

Every signal is built twice from the same events: a *physical* basis that uses hindsight, for validation, and a *knowable* basis that at each day d uses only what could have been on a screen on day d, for modelling. The distinction is not cosmetic. Two properties of voyage-derived features make the knowable basis hard to get right, and both were found the hard way.

3.1 The unit of observation is defined retrospectively

A voyage leg does not exist until both a departure and the next arrival have been observed. At time t a vessel that departed five days earlier has an *incomplete* leg of unknown duration and unknown destination; in hindsight it has a complete leg with both. Any statistic over "legs" therefore conditions on completion unless the population is explicitly bounded. Figure 2 quantifies the effect: at an as-of date of 1 June 2020, a loader that pairs all departures with all arrivals and then filters by date returns 20,351 legs, of which 13,130 (65%) are closed by arrivals that had not yet happened. Bounded correctly, 4,713 legs remain, none closed by the future. Status labels move with them: the count of legs classified as "floating" fell from 95 to 9 under the bound, because that label is assigned in hindsight. The correction is a strict reduction — at an as-of date past the data maximum both paths agree on every count — which is the test that it is a bound and not a different query.

3.2 Feature availability changes across vintages

The same naive loader attached 2,101 destination declarations to the 2020 legs. Declarations come from a live AIS static-data feed that began in 2026; no declaration existed in 2020. This

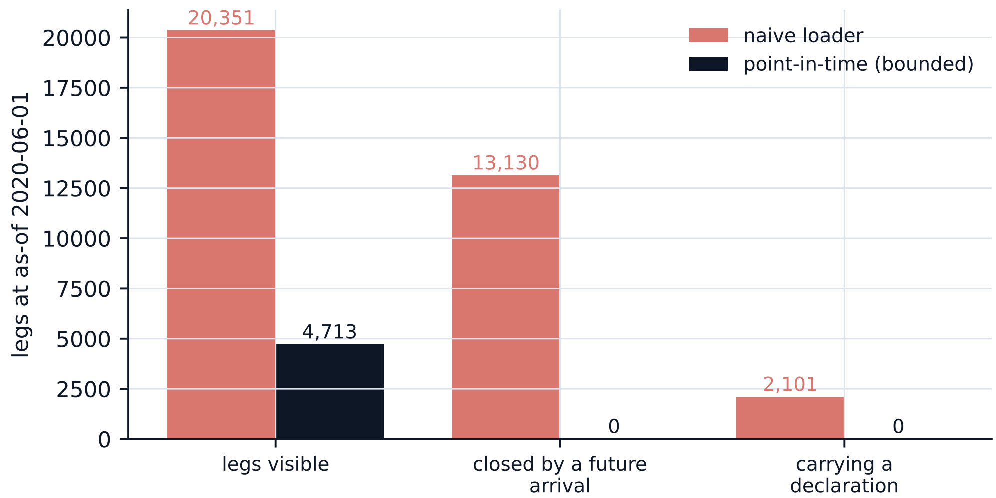

Figure 2: Voyage legs visible at an as-of date of 1 June 2020 under the naive loader and under the point-in-time bound. Recomputed live from the event tables for this figure.

is a different failure from the first: a date filter on the *observations* is satisfied, because each declaration row has a timestamp in the past of the vessel's current voyage, yet the *table* being joined should not exist at that vintage. The bound therefore substitutes a reconstruction of the declaration from the raw static messages available at d, and at a 2020 as-of date it returns none.

3.3 What "knowable" bounds, and what it does not

The knowable basis bounds on event time. Event timestamps are back-dated to the fix at which a transition first qualified, so a departure is dated to the vessel's last fix in the berth polygon rather than to the later fix that confirmed it had gone. The gap between the two is the feature's own publication lag. We measured it directly: the first fix after a departure event arrives at a median lag of zero hours, and more than 24 hours later for 0.3% of departures on the NOAA feed (n = 1,500) and 1.5% on the live feed (n = 343). On the decade panel this is immaterial.

A separate log records the knowable value of every signal as it was printed each day of live operation, so that the bound can be checked rather than asserted. Comparing the rebuilt knowable series with those prints on days old enough to have settled, the check flags 68 rows whose value moved after a next-day print, all in the live tail, consistent with late-arriving fixes back-dating departures. (A further 909 rows differ from prints taken *during* the day, before that day's fixes had finished arriving; that comparison measures intraday timing, not leakage, and is reported but not scored.) The live tail lies outside the modelling sample used in Section 6.

3.4 The modelling grid

Prices, storage and the tanker signals arrive on different calendars. They are assembled onto one daily grid by forward-fill under a per-cadence staleness ceiling, never by an inner join — Henry Hub spot prints on 69% of calendar days, and an inner join across the price legs would silently drop a third of the history. Each external series is shifted to the date it became public (EIA weekly storage +6 days, daily spot +1, AGSI+ +1); the tanker signals are shifted by zero, because the knowable basis is already point-in-time and shifting again would double-count. Degree days are deliberately *not* shifted: their role is to be partialled out, so what

Table 3: Pre-registration entries. The commit column is the honest part: see the text.

entry	dated	what was fixed before the fit	commit
D-001–D-003	2026-08-10	A1 target, kernel, replay protocol	`4239d13`
D-016	2026-08-12	A2 design lock	`4239d13`
D-019	2026-08-12	A5 design lock	`4239d13`
D-022	2026-08-12	A4 design lock	`4239d13`
D-025	2026-08-12	H1 horizon test pre-specified	`4239d13`
D-028	2026-09-05	Part B null + FWL: target, controls, HAC lag, signs	`e483a1b`
D-031	2026-09-05	H2/H3/H4: signs, Bonferroni bar, holdout	`c489550`

matters is the weather that happened, not what was knowable. Signals reported per terminal or per destination zone are collapsed to one series by summing volumes and by n-weighted averaging of durations.

3.5 The coverage seam

The two backfill feeds end on 2025-12-31 (NOAA) and 2026-02-21 (GFW). After that the signals are carried by the live feed alone — 3,238 events against 84,636 — and the EU-bound at-sea stock falls from 4.80 to 0.41 million m^3 between December 2025 and January 2026, a 91% drop with no physical event behind it. This was found by plotting Figure 1, not by any test. The primary modelling sample is therefore truncated at 2025-12-31; the full-panel results are reported in Appendix B and change no conclusion.

4 Pre-registration protocol

The project keeps a dated, append-only decision log. A hypothesis, target, lag, sign or acceptance bar is committed the day its entry lands, before the corresponding fit; a later change is a visibly dated superseding entry that names the result which prompted it. The log is the instrument that turns a null into a finding: without it, a reader cannot distinguish "we tested and found nothing" from "we kept looking until nothing crossed a line, and reported the line". Table 3 lists the entries that fixed each model below.

Two consequences of the protocol are visible in the results. A pre-registered sign falsified by the data is reported as falsified (Section 5); and a test that produced an apparent 87% improvement over its null was withdrawn, because the protocol's own comparability condition failed (Section 5). Neither would have happened without a committed criterion to fail against.

What the log does not establish. The log is an ordinary text file in the project's version control, and its git history does not independently prove the ordering it asserts. The entries for the physical nowcasts (D-001 through D-026) were committed together, weeks after the work they describe; the two spread pre-registrations (D-028 and D-031) were each committed in the same change as their result. The *substance* of the pre-registration — the signs, bars, horizon and holdout fixed in each entry's text — was written before the corresponding run, but a sceptical reader has only the author's word for it. We state this rather than have it discovered, and note that a third-party timestamped registry would have cost nothing. Future entries are committed and pushed before the run they govern.

Table 4: One-week-ahead nowcasts of weekly US loadings (A2, A4) and conditional EU arrivals (A1), scored walk-forward from 2018 on the NOAA feed. MAE in cargoes per week. All three fail their pre-registered bar of beating both nulls.

model	MAE	persistence	climatology	vs climatology	80% cov.
A1 arrival-count climatology	2.314	2.217	1.814	+27.5%	73.7%
A2 negative-binomial GLM	3.961	2.551	2.152	+84.0%	75.7%
A4 Kalman local level	2.104	2.694	2.088	+0.7%	72.6%

Table 5: Outage detection against labelled terminal outages. N2 is the pre-registered null; it is the usable artefact, and its recall is the number to read.

detector	recall	median delay (d)	false alarms / terminal-yr
BOCPD (pre-registered)	12% (2/16)	17	0.13
N1 absolute silence (14 d)	88% (14/16)	14	2.33
N2 rate-relative silence	38% (6/16)	12	0.19

5 Physical nowcasts: the naive mean is near-optimal

Before asking whether the signals predict a price, we asked whether they predict the physical quantities they directly measure — weekly EU arrivals, weekly US loadings, terminal outages — at one- and two-week horizons. These are high signal-to-noise targets with published ground truth, and they establish what the pipeline can and cannot see before any price enters. Four models were pre-registered and scored walk-forward against two naive nulls (persistence and a four-week climatology), each built to fix the diagnosed weakness of the last. Table 4 gives the one-week headline.

The sequence is the result. A1, a duration-climatology convolution of the at-sea stock, is +28% worse than the climatology null because its estimator is maturity-gated and stale. A2, a negative-binomial count model on lagged loadings and berth occupancy, is +84.0% worse because it over-extrapolates the trend; it also falsified one of its pre-registered signs (ballast arrivals entered with coefficient -0.049 against a prior of positive), and this is reported as falsified. A4, a Kalman local-level model, is +0.7% from the null — a tie. The tie is informative rather than disappointing: A4 fitted its smoothing constant by maximum likelihood and chose $\alpha = 0.251$, an effective window of 7.0 weeks. For a series that is a local level plus noise, the exponentially weighted moving average is the optimal linear predictor [8]; the fit landing on top of a four-week simple average is the optimum being found, not four methods coincidentally tying. On this target, at this horizon, the naive mean is near-optimal, and a fifth method would be wasted effort.

Outage detection. A5 tested Bayesian online change-point detection [1] on the inter-departure process at each US terminal against 16 labelled outages over 54 terminal-years (Table 5). It did not beat a rate-relative silence rule, which detects with a median delay of 12 days at 0.19 false alarms per terminal-year — but with a recall of only 38% (6/16). The rule is a usable, limited monitor, and it is the baseline, not the model.

A test withdrawn. The natural next step was to ask whether skill rises with horizon — cargo at sea today determines arrivals two to three weeks out — and this was pre-specified before any longer-horizon data was examined. Extending A1's window to four weeks produced a +86.7% improvement over the best null at week four (Table 6). It is an artefact. A1's target is *conditional* on legs already at sea, while both nulls are the last elapsed *unconditional* weekly

Table 6: A1 extended to four weekly windows. The "skill" column at W3 and W4 is an artefact of a conditional target scored against unconditional nulls; the harness now refuses to report it as a result.

window	mean truth	A1 MAE	persistence	climatology	"skill"	80% cov.
W1	5.89	2.324	2.226	1.822	−27.5%	73.5%
W2	5.49	2.121	2.052	1.773	−19.6%	76.0%
W3	1.98	1.170	4.146	4.058	+71.2%	85.9%
W4	0.75	0.691	5.247	5.208	+86.7%	94.2%

count. The two coincide at one and two weeks — no US-to-EU voyage completes in under seven days — but past the voyage tail the conditional truth empties toward zero (mean 5.89 arrivals in week one, 0.75 in week four) while the nulls keep predicting the full weekly rate (5.25 MAE). A1 was not forecasting better; it was forecasting a smaller quantity. The pre-specified premise — that the nulls degrade with horizon — fails, and the protocol said that outcome means abandon rather than patch. In the comparable region, skill moves from −27.5% to −19.6%: the predicted direction, but from far below the null to clearly below it.

6 The spread

6.1 Target, grid and controls

The target is the spread $s_t = \mathrm{HH}_t - \mathrm{TTF}_t$ in \$/MMBtu, with TTF converted from EUR/MWh at the day's EUR/USD rate and 3.412 MWh per MMBtu, sampled weekly on Mondays from 2018 (the same grid as the physical nowcasts). The scored quantity is the h-week-ahead change, $\Delta_h = s_{t+7h} - s_t$, at $h \in \{1, 4\}$; the level is too persistent to score honestly, since a model that predicts "about the same as now" looks skilled on it. The primary sample runs to 2025-12-31 (Section 3), giving 379 complete weeks.

The control block is the spread's non-tanker drivers: heating and cooling degree days for the US and north-west Europe, US and EU storage, Brent, and a winter dummy. The price legs that construct the target are excluded as regressors. Every standard error is Newey–West [9] with the lag fixed in advance at $h + 1$ weeks: on a weekly grid at $h = 4$, consecutive targets share three weeks and an ordinary least-squares standard error is roughly $\sqrt{h}$ too small. Fits are walk-forward with an expanding window and a 104-week minimum, and a training row is admissible only if its target closed strictly before the test week, plus a one-week embargo [6].

6.2 The controls-only null is worse than no change

Three models form a ladder, each the null for the next: M0 predicts no change; M1 is an AR(1) in the level; M2 adds the controls. M2 is the "AR(1)+controls" null the tanker signals were pre-registered to beat. Table 7 shows that it does not beat M0: relative to assuming no change, M2 scores −9.3% at one week and −18.2% at four, monotonically worse at the longer horizon, and worse in each of 2023, 2024 and 2025 individually. This is what an efficient, persistent price series looks like — the level already embeds the fundamentals, so re-adding them as regressors imports estimation noise — and it means the operative null for the signals is M0, a harder bar than the one pre-registered.

Table 7: The model ladder on the primary sample. Skill is the fractional MAE improvement over M0; negative means worse than assuming no change.

h (wk)	model	n	MAE	RMSE	80% cov.	skill vs M0
1	M0	272	2.034	4.465	81.6%	–
1	M1	272	2.061	4.571	81.6%	−1.4%
1	M2	272	2.223	4.630	82.4%	−9.3%
4	M0	266	3.942	7.203	72.9%	–
4	M1	266	4.196	7.585	70.7%	−6.4%
4	M2	266	4.660	7.437	69.5%	−18.2%

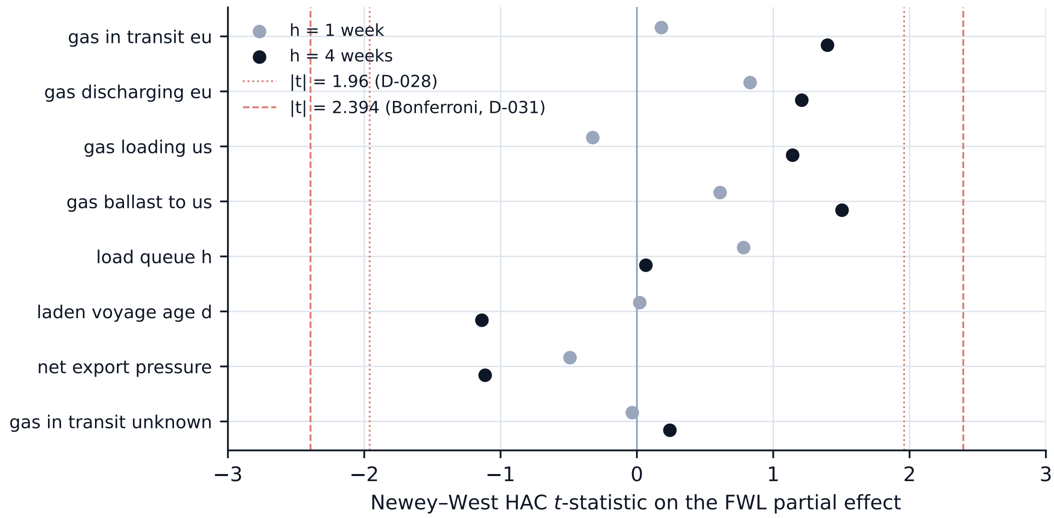

Figure 3: Newey–West t-statistics on the FWL partial effect of each pre-registered signal, net of persistence, weather, storage and oil, at one and four weeks. The dotted and dashed lines are the two decision bars used in the paper.

6.3 No signal survives partialling

For each of eight pre-registered signals we residualise both the target and the signal on M2's regressors and regress residual on residual — the Frisch–Waugh–Lovell partial effect [3, 7], which by the theorem equals the signal's coefficient in the full regression and is the harness's own correctness test. Each signal carried a sign committed in advance (Appendix A), and a result counted only if $|t| > 1.96$, the sign matched, and the sign held in at least two-thirds of individual years — the last condition guarding against 2022 alone carrying a pooled fit.

None does (Figure 3, Table 8). The largest $|t|$ at either horizon is 1.50; the largest partial R^2 is 0.009. The cleanest mechanical story — cargo at sea today lands in Europe in two to three weeks — was tested directly as the EU-bound stock and gave $t = +0.18$ at one week and $+1.40$ at four. At four weeks the four flow signals all carry their pre-registered sign, comfortably inside the noise band: directionally consistent with the mechanism and uniformly insignificant. No sign was falsified, because nothing reached significance to falsify.

6.4 Three mechanisms a linear scan could miss

A one-at-a-time linear scan has three specific blind spots, and each was tested with a feature or a split rather than a new model class — so each costs one test. The three were pre-registered together with a Bonferroni bar of $|t| > 2.394$, a primary horizon of four weeks chosen on the

Table 8: FWL partial effects on the primary sample. Year consistency is the share of individual years whose partial effect carries the full-sample sign.

		$h = 1$		$h = 4$		
signal	prior	HAC t	partial R^2	HAC t	partial R^2	year cons.
gas in transit, EU-bound	+	+0.18	0.000	+1.40	0.009	62%
gas in transit, unresolved	none	-0.03	0.000	+0.24	0.000	50%
US loading rate	+	-0.32	0.000	+1.14	0.006	62%
EU discharge rate	+	+0.83	0.001	+1.21	0.003	62%
ballast returning to US	+	+0.61	0.001	+1.50	0.008	62%
laden voyage age	−	+0.02	0.000	-1.14	0.004	62%
US load-queue wait	−	+0.78	0.001	+0.07	0.000	38%
net export pressure	+	-0.49	0.000	-1.11	0.003	75%

Table 9: The three pre-registered mechanism tests, discovery sample, $h = 4$, Bonferroni bar $|t| > 2.394$. None is significant; H3 carries the wrong sign; H4's tight-regime effect is smaller than its loose-regime effect.

hypothesis	prior	$\hat{\beta}$ (discovery)	HAC t	year cons.	partial R^2	$\hat{\beta}$ (holdout)
H2: EU share of at-sea stock	+	-2.64×10^{0}	-0.89	71%	0.003	-2.51×10^{0}
H3: transit × EU storage	−	4.78×10^{-9}	+0.46	71%	0.001	-1.49×10^{-9}
H4: transit, tight regime	+	-1.13×10^{-7}	-0.17	100%	0.000	2.71×10^{-7}
H4: transit, loose regime	+	-2.25×10^{-7}	-1.00	100%	0.009	–

voyage mechanism, and a holdout (2025-01-06 to 2025-12-01, 48 weeks) not examined until a hypothesis had passed on discovery (2018-01-01 to 2024-12-30, 327 weeks). A pass required the holdout sign to replicate.

H2: the share, not the level. Most of the at-sea stock has no resolved destination, so two collinear levels with offsetting effects could each look like noise while their ratio carries the signal. A pre-registered gate voids the test if the share is mostly a decade-long ramp (coverage improving rather than cargo re-routing); the trend explains $R^2 = 0.209$ and the gate passes. *H3: an interaction with EU storage.* Extra cargo landing at 90% full does little; the same cargo at 30% is highly price-relevant, and a linear fit averages the two worlds. *H4: a tightness split.* The cheap form of a regime model: the scan re-run within weeks where EU storage sits below its seasonal norm (norm fitted on discovery only), with the prediction that the effect is positive there and larger than in loose weeks.

All three fail (Table 9), and none is close: the largest $|t|$ among the primary tests is 0.89. Each fails in a way that argues against another attempt. H2's estimate is stable and consistently negative — the wrong sign, on discovery (-2.64×10^{0}) and holdout (-2.51×10^{0}) alike — and never near significance. H3 comes out positive ($t = +0.46$) against a prior of negative, so neither the magnitude nor the direction of the storage interaction is present. H4's tight-regime effect is negative ($t = -0.17$) and smaller in magnitude than the loose-regime effect, the reverse of both predictions. Cargo arrivals do not matter more when Europe is short; on this evidence they do not measurably matter in either state.

Taken together, the linear null of the previous subsection has been tested against the three specific reasons it could have been wrong, on pre-registered signs, with a corrected bar and an untouched holdout, and it stands.

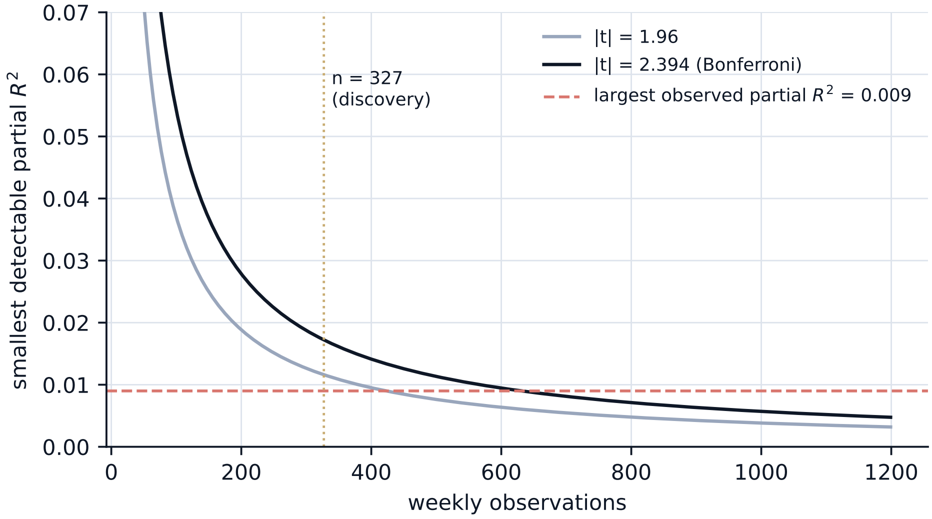

Figure 4: Smallest detectable partial R^2 against sample size at the two bars used in the paper, with the largest observed partial R^2 and the discovery sample size marked.

7 What the design could have found

A null is only as informative as the smallest effect it could have detected. For a partial effect estimated on n observations, $t^2 \approx n\,R^2/(1 - R^2)$, so at a fixed bar the smallest detectable partial R^2 falls as $1/n$. With 327 discovery weeks and the Bonferroni bar of $|t| > 2.394$, the floor is 1.72% of residual variance (Figure 4). The largest partial R^2 observed anywhere in the paper — across eight signals, two horizons, three mechanisms and two samples — is 0.009.

The scope of the result follows. It rules out effects large enough to be worth trading and is silent on smaller ones: a signal explaining a third of a percent of residual variance would be real physics and a worthless strategy, and this design could not distinguish it from zero. Two further scale facts make the null less surprising than the intuition that motivated the study. The entire EU-bound stock of gas at sea averages 3.69 million m^3 of LNG, about 2.21 bcm of gas or 2.3 days of EU demand at roughly 350 bcm per year; a one-standard-deviation swing in it is 1.37 days of demand. The weekly spread, meanwhile, has a standard deviation of \$11.22 per MMBtu. The question was whether a day and a half of marginal supply moves a price that routinely swings on a revised weather forecast.

8 Discussion

The null is the answer an efficient market predicts, and it is worth being precise about which question was asked. LNG cargo tracking is a commercial industry: several vendors sell satellite-augmented AIS with proprietary destination inference to every gas trading desk, in real time. The signals in this paper are built from public terrestrial AIS and a free voyage archive — a strictly coarser version of what the marginal trader already has. The test was therefore not "is the physical signal informative" but "is public physical data mispriced in a liquid benchmark at one- to four-week horizons", and to that question a negative answer is close to by construction. What the paper adds is that the negative is not an artefact of a broken pipeline, an unlucky specification, or a search that stopped at the first $t > 2$: the pipeline is validated against a published total, the specification was fixed in advance, and the search was corrected and held out.

Three results deserve to survive the null. First, the physical nowcasts converge on the naive mean from above, and for the one-week loadings target this is a theorem rather than an observation — an argument that applies to any slow industrial flow series and should temper the expectation that "more model" will beat a moving average on one. Second, the two construction hazards of Section 3 are general to voyage-derived features and easy to miss while feeling that dates have been handled correctly; the vintage log that catches them is cheap to keep and, as far as we know, rarely kept. Third, the outage rule of Section 5 is a usable monitor with a stated recall.

What would change the answer? A private edge — satellite coverage of the open ocean, a destination model that resolves the three-quarters of the stock this pipeline cannot — would change the question rather than the answer, and would make the study a test of the vendor's product. Within public data, the remaining untested avenues are joint rather than marginal fits (spike-and-slab or partial least squares over the collinear block), a fitted regime model rather than an observable split, distributed lags of the signals, and the twenty-six signal keys the panel does not use. Each is a weaker prior than the three mechanisms already tested, and each faces a worse multiplicity problem against a residual that has never exceeded 0.009 of variance. Any of them should arrive with its own pre-registration, its own correction, and the untouched-holdout standard used here.

9 Limitations

The European side has no anchorage history. The GFW feed carries port visits but no anchorage events, so European queue and absorption signals exist only from the 2026 live feed. Four of the five composite signals originally pre-registered as Part B features turned out to have between 9 and 46 daily observations and were excluded by a documented amendment before any spread fit. The at-sea and US-side mechanism was tested on the decade; the EU-congestion mechanism was not testable.

Destination blindness. Between 75% and 83% of the at-sea stock has no resolved destination. The EU-bound series is a minority observed slice whose definition depends on coverage; the unresolved band is carried as its own feature, and the ratio of the two was tested as H2. Attenuation from this measurement error pushes coefficients toward zero and cannot be excluded as a contributor.

The capture gradient. NOAA-only capture of US export cargoes is 46% in 2020 and 74% in 2018 (Table 2). Early-year levels of every US signal are biased low. The walk-forward designs work on within-window changes and the FWL scan on residuals, but a level-dependent non-linearity in the early years would be misread.

The coverage seam. The backfill feeds end on 2025-12-31 and 2026-02-21. The full-panel results in Appendix B include the thin live tail; 24% of that sample's holdout weeks sit past the seam, and one full-panel finding (a holdout sign flip on H2) did not survive truncation and is not claimed. The primary sample is truncated at the seam.

Regime pooling. Signals are read from the pooled regime tag, the only one spanning the decade. A small residual over-count where NOAA and GFW both see the same US departure is documented in the project's signal catalogue; a robustness run on the NOAA tag alone for US-side signals was not performed.

The pre-registration trail. As Section 4 states, git does not independently establish that the log entries preceded the fits they govern.

Sample size and one dominant regime. The scored spread sample is 379 weeks, heavily autocorrelated, and 2022 alone spans a range larger than the rest of the decade combined. The year-consistency condition and the holdout are the defences; a longer clean sample would be better than either.

Vintage residual. 68 rows of the live tail moved after a next-day print (Section 3). They lie outside the modelling sample, but they show that the knowable bound is exact on event time and only approximately exact on observation time in a thin-coverage regime.

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A Signal definitions and pre-registered signs

The eight tanker signals used in Section 6, the aggregation applied when collapsing per-terminal or per-zone bands to one daily series, and the sign committed before any fit. The sign convention: the spread HH − TTF is negative in nearly every period, so "the spread narrows" means it rises toward zero — a positive change.

signal	aggregation	mechanism	prior
gas in transit, EU-bound	sum over EU zones	more cargo landing in Europe soon ⇒ TTF falls	+
gas in transit, unresolved	sum	destination unknown; no directional prior	none
US loading rate	sum over US berths	gas leaves the US balance (HH firms) and reaches Europe (TTF falls)	+
EU discharge rate	sum over EU berths	Europe absorbing gas now ⇒ TTF falls	+
ballast returning to US	sum	tonnage returning to load ⇒ future exports up	+
laden voyage age	n-weighted mean	cargo floating rather than landing ⇒ Europe stays tight	−
US load-queue wait	n-weighted mean	US cannot export ⇒ HH softens, Europe starved	−
net export pressure	n-weighted mean	z(loadings) − z(load queue); both legs point the same way	+

Volumes are in m^3 of LNG; loading and discharge rates are amortised m^3 per day across berth hours so that a closed visit integrates to one cargo; ages and waits are in days and hours respectively. The project's signal catalogue documents the remaining twenty-six keys, the two bases, and the confidence columns (dispersion, open fraction, estimated fraction) that accompany every value.

A recorded inconsistency. The catalogue's own text gave net export pressure two incompatible signs in two places. The positive sign was adopted on the economics above and recorded as a prior; the data returned a negative point estimate at both horizons, at $|t| \leq 1.6$, which neither confirms nor falsifies it.

B Full-panel results

The primary sample is truncated at the backfill horizon (2025-12-31). The same scans run to the data maximum (395 weeks) are reported here for completeness; they were the runs first published in the decision log, and they change no conclusion.

C Reproducibility, code, and the use of AI

Reproducibility. Every number in this paper is generated. The replays write machine-readable results (`make paper-results`); a generator turns them into the LaTeX tables and into a file of macros from which every in-prose number is drawn (`make paper-tables`); the

Table 10: Model ladder, full panel.

h (wk)	model	n	MAE	RMSE	80% cov.	skill vs M0
1	M0	288	1.997	4.371	82.3%	–
1	M1	288	2.023	4.473	82.3%	−1.3%
1	M2	288	2.205	4.542	83.0%	−10.4%
4	M0	282	3.947	7.090	73.8%	–
4	M1	282	4.168	7.444	72.0%	−5.6%
4	M2	282	4.705	7.378	69.1%	−19.2%

Table 11: FWL partial effects, full panel. The largest $|t|$ here (1.72) belongs to the unresolved-destination band and falls to +0.24 on the primary sample: it was the contaminated tail.

		$h = 1$		$h = 4$		
signal	prior	HAC t	partial R^2	HAC t	partial R^2	year cons.
gas in transit, EU-bound	+	+0.08	0.000	+1.03	0.004	75%
gas in transit, unresolved	none	-0.74	0.001	-1.72	0.009	62%
US loading rate	+	-0.19	0.000	+0.63	0.001	62%
EU discharge rate	+	+0.84	0.001	+0.46	0.000	62%
ballast returning to US	+	+0.14	0.000	+0.31	0.000	62%
laden voyage age	−	-0.04	0.000	-1.26	0.005	62%
US load-queue wait	−	+1.01	0.001	+0.55	0.001	38%
net export pressure	+	-0.82	0.001	-1.60	0.009	75%

figures are drawn from the same tables the models read (`make paper-figures`). A number that is not generated cannot appear, because the macro is undefined and the document does not compile. The underlying replays are `make a1-replay`, `a2-replay`, `a4-replay`, `a5-replay`, `a1-h1`, `b0-coverage`, `mechanisms-coverage` and their full-panel variants; the panel itself is `make signals` followed by `make model-panel`. The source, the decision log and the test suite (591 tests at the time of writing) are public at the project repository.

Data and licences. AIS positions are from NOAA's Marine Cadastre and Global Fishing Watch under their respective terms; EIA, FRED, GIE AGSI+ and Open-Meteo are free public APIs; the World Bank monthly series is CC BY 4.0 and is the only price series redistributed with the code. The daily TTF series is obtained from a public quote feed under personal-use terms and is not redistributed; the loader re-fetches it.

Use of AI. The author used an AI coding assistant (Anthropic's Claude, via Claude Code) extensively in this project. A substantial fraction of the pipeline, the modelling harnesses, the tests and the first drafts of this text were written with its assistance. Every design decision, hypothesis, sign and acceptance bar in the decision log is the author's; every result was reproduced by the author from the stated `make` targets; and the author takes full responsibility for the content. The assistant was also used to audit the project's own claims before this paper was written, which produced the corrections recorded in the decision log's final entries.

Table 12: Mechanism tests, full panel. The holdout coefficient for H2 changes sign relative to the primary sample; 24% of this holdout's weeks lie past the coverage seam, and the sign flip is not claimed.

hypothesis	prior	$\hat{\beta}$ (discovery)	HAC t	year cons.	partial R^2	$\hat{\beta}$ (holdout)
H2: EU share of at-sea stock	+	-2.64×10^{0}	-0.89	71%	0.003	8.23×10^{0}
H3: transit × EU storage	−	4.78×10^{-9}	+0.46	71%	0.001	6.73×10^{-10}
H4: transit, tight regime	+	-1.13×10^{-7}	-0.17	100%	0.000	9.21×10^{-7}
H4: transit, loose regime	+	-2.25×10^{-7}	-1.00	100%	0.009	–

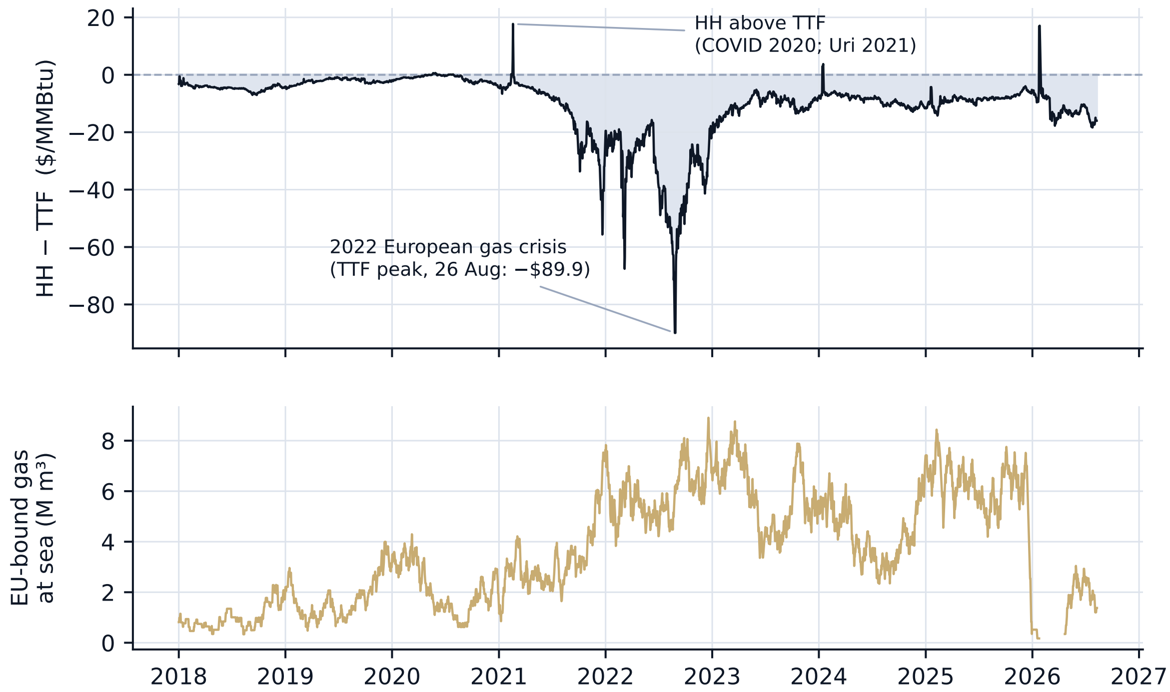

Figure 5: Figure 1 extended to the data maximum. The 91% step in the at-sea stock at the start of 2026 is the coverage seam, not a physical event.

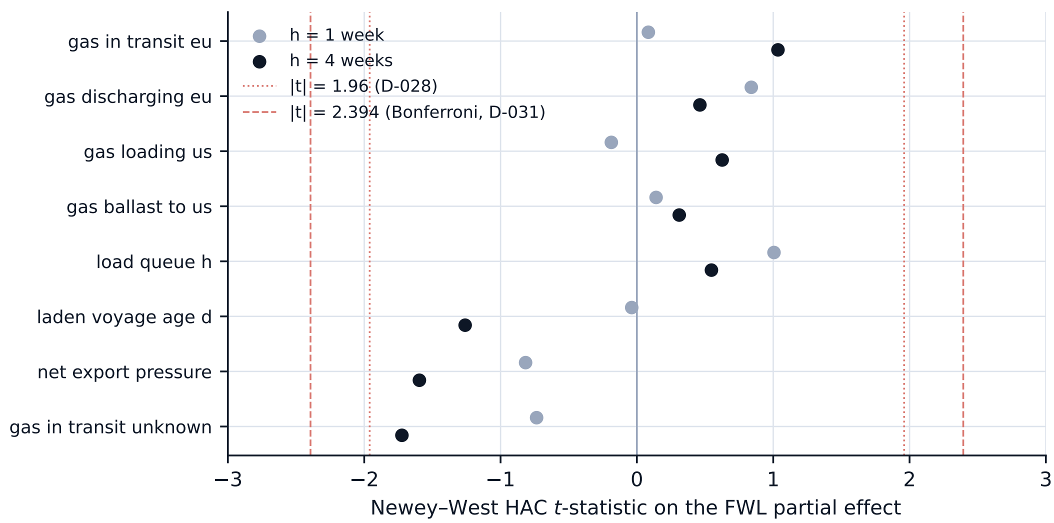

Figure 6: Figure 3 on the full panel.